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SRI RAMAKRISHNA ENGINEERING COLLEGE, COIMBATORE

(Autonomous Institution, Reaccredited by NAAC with 'A+' Grade Approved by AICTE and Permanently affiliated to Anna University, Chennai)

AUTONOMOUS EXAMINATIONS - MAY'2025

COMMON TO B.E – CSE / B.TECH – IT / AIDS & M.TECH - CSE

20MA214 - DISCRETE STRUCTURES AND COMBINATORICS

Answer ALL questions

Duration: 3 Hours

Maximum: 100 Marks

PART - A

(7x 1 = 7 Marks)

1. CO2 The relation R on {1, 2, 3, 4} by $R = \{(a, b) / |a - b| = 1\}$ is _____.
a) reflexive b) symmetric c) anti-symmetric d) transitive
2. CO2 A binary relation R in a set P is called a partial order relation in P if and only if R is _____.
a) reflexive and transitive b) reflexive and antisymmetric
c) transitive and antisymmetric d) reflexive, antisymmetric and transitive
3. CO1 In any Boolean algebra, $(a + b)(a' + c) =$ _____.
a) $ac + a'b + ab$ b) $ac + a'b + bc$ c) $ab + a'c + ac$ d) $bc + a'c + ac$
4. CO3 A set of premises $p_1, p_2, p_3, \dots, p_n$ is said to be inconsistent if their conjunction implies _____.
a) tautology b) neither tautology nor contradiction
c) both tautology and contradiction d) contradiction
5. CO3 If p and q are any two statements, then the statement $p \leftrightarrow q$ is called _____ statement.
a) conditional b) conjunction c) bi-conditional d) disjunction
6. CO3 The proposition which is equivalent to $P \rightarrow (Q \rightarrow R)$ is _____.
a) $(P \wedge Q) \rightarrow R$ b) $(P \vee Q) \rightarrow R$ c) $\neg(P \wedge Q) \rightarrow R$ d) $\neg((P \wedge Q) \rightarrow R)$

7. CO4 To solve the recurrence relation $a_{n+1} - 2a_n = 3n^2 - n$, the trial particular solution is assumed to be _____.
 a) An^2 b) $An - 2$ c) $An^2 + Bn$ d) $An^2 + Bn + C$

Fill in the blanks:

(3x 1 = 3 Marks)

8. CO2 A relation R is an equivalence relation if R is _____.
 9. CO4 An ordered arrangement of r elements of a set containing n distinct elements is called _____.
 10. CO4 The number of different circular arrangements of n objects is _____.

PART - B

(5x 3 = 15 Marks)

11. CO2 List the elements of relation R on the set $A = \{1, 2, 3, 4\}$ defined by the rule $R = \{(a, b) / |a - b| = 1\}$.
 12. CO2 If $A = \{1, 2, 3, 4, 5\}$, $B = \{1, 2, 3, 8, 9\}$ and the functions $f: A \rightarrow B$, $g: A \rightarrow A$ are defined by $f = \{(1, 8), (4, 3), (3, 9), (2, 1), (5, 2)\}$ and $g = \{(1, 2), (2, 2), (3, 1), (4, 3), (5, 2)\}$, find $f \circ g$ and $g \circ g$.
 13. CO3 Construct a truth table for the following compound proposition: $(p \vee q) \rightarrow (p \wedge q)$
 14. CO4 Find the recurrence relation for the Fibonacci sequence.
 15. CO4 Write down the disjoint cycles of the permutation function $f = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 5 & 1 & 2 & 4 & 6 \end{pmatrix}$.

Also classify f as even or odd permutation.

PART - C

(5 x 15 = 75 Marks)

Q.No: 16 Compulsory Question

Q.No: 17 to 22 Answer any FOUR Questions

Compulsory Question:

16. CO2 i) If R is the relation on the set of integers such that $(a, b) \in R$ if and only if $3a + 4b = 7n$ for some integer n, prove that R is an equivalence relation. (8)
 CO2 ii) Draw the Hasse diagram representing the partial ordering $\{(A, B) : A \subseteq B\}$ (7)
 on the power set P(S) where $S = \{a, b, c\}$. Find the maximal, minimal, greatest and least elements of the poset.

17. **CO1** i) In a distributive lattice, show that $(a * b) \oplus (b * c) \oplus (c * a) =$ (7)

$$(a \oplus b) * (b \oplus c) * (c \oplus a)$$

CO2 ii) Let $A = \{1, 2, 3\}$ and R, S be defined on A such that (8)

$$M_R = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad M_S = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

Find the matrices of (1) $(R \cup S)^C$ (2) $(R \cap S)^{-1}$ (3) $(R \circ S)^{-1}$ (4) R^2 (5) $R \oplus S$

18. **CO2** i) If $A = \left\{x \in R : x \neq \frac{1}{2}\right\}$ and $f : A \rightarrow R$ is defined by $f(x) = \frac{4x}{2x-1}$, (7)

- find the range of f
- Check whether f^{-1} exists. If so find the domain and range of f^{-1} .

CO3 ii) Prove that conclusion $\forall x(P(x) \rightarrow \neg Q(x))$ follows from the premises (8)

$$\exists x(P(x) \wedge Q(x)) \rightarrow \forall y(R(y) \rightarrow S(y)) \text{ and } \exists y(R(y) \wedge \neg S(y))$$

19. **CO3** i) Determine whether the conclusion C follows logically from the premises (7)

$$H_1 \text{ and } H_2 \quad H_1 : P \rightarrow Q, \quad H_2 : \neg(P \wedge Q) \quad C : \neg P$$

CO3 ii) Check whether the proposition $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$ is a (8)
tautology or contradiction

20. **CO3** i) Obtain the principle disjunctive and principle conjunctive normal forms of (8)

$$(p \vee q) \wedge (r \vee \neg p) \wedge (q \vee \neg r).$$

CO3 ii) Show that $\forall x(P(x) \rightarrow (Q(y) \wedge R(x)))$, $\exists x P(x) \Rightarrow Q(y) \wedge \exists x(P(x) \wedge R(x))$ (7)

21. **CO4** i) A man hiked for 10 hours and covered a total distance of 45 km. It is (7)

known that he hiked 6 km in the first hour and only 3 km in the last hour. Show that he must have hiked at least 9 km within a certain period of 2 consecutive hours.

CO4 ii) Find the number of integers between 1 and 250 both inclusive that are not (8)
divisible by any of the integers 2, 3, 5 and 7

22. **CO4** i) Solve the recurrence relation : $a_{n+1} - 2a_n = 5, n \geq 0, a_0 = 1$ (8)

CO4 ii) A survey of 500 television watchers produced the following information: (7)

285 of them watch football games, 195 watch hockey games, 115 watch basketball games, 70 watch football and hockey games, 50 watch hockey and basketball games, 45 watch football and basketball games 50 do not any of the three games. How many people watch exactly one of the three games?
